

Problem

Determine the Fourier transforms of these functions: (a) gate function $g(t) = u(t) - u(t - 1)$, (b) $f(t) = 10t - 2tu(t)$, and (c) sawtooth pulse $p(t) = 75t[u(t) - u(t - 2)]$.

Step-by-step solution

Step 1 of 5

(a)

The function is,

$$g(t) = u(t) - u(t - 1)$$

The signal $u(t)$ is expressed as,

$$u(t) = \frac{1}{2} + \frac{1}{2} \operatorname{sgn}(t)$$

The Fourier transform of 1 and $\operatorname{sgn}(t)$ are $2\pi\delta(\omega)$ and $\frac{2}{j\omega}$ respectively. Hence,

$$\begin{aligned} F[u(t)] &= \frac{1}{2} \left[2\pi\delta(\omega) + \frac{2}{j\omega} \right] \\ &= \pi\delta(\omega) + \frac{1}{j\omega} \end{aligned}$$

Applying Time shifting property,

$$\begin{aligned} F[u(t - 1)] &= e^{-j\omega} F[u(t)] \\ &= e^{-j\omega} \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] \end{aligned}$$

Step 2 of 5

The Fourier transform of the function $g(t)$ is,

$$\begin{aligned} G(\omega) &= \int_{-\infty}^{\infty} [g(t)] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} [u(t) - u(t - 1)] e^{-j\omega t} dt \end{aligned}$$

$$G(\omega) = F[u(t)] - F[u(t - 1)]$$

Substitute $\left[\pi\delta(\omega) + \frac{1}{j\omega} \right]$ for $F[u(t)]$ and $e^{-j\omega} \left[\pi\delta(\omega) + \frac{1}{j\omega} \right]$ for $F[u(t - 1)]$.

$$\begin{aligned} G(\omega) &= \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] - e^{-j\omega} \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] \\ &= (1 - e^{-j\omega}) \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] \end{aligned}$$

Thus, the Fourier transform of the function $g(t)$ is $\boxed{(1 - e^{-j\omega}) \left[\pi\delta(\omega) + \frac{1}{j\omega} \right]}$.

Step 3 of 5

(b)

The function given is wrong and the Fourier Transform cannot be calculated. Hence consider the function to be in the standard form as,

$$f(t) = 10te^{-2t}u(t)$$

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} 10te^{-2t}u(t) dt \\ &= 10 \int_{-\infty}^{\infty} te^{-2t}u(t) dt \end{aligned}$$

The standard Fourier Transform pair is, $t^n e^{-at}u(t) \xleftrightarrow{F} \frac{n!}{(a + j\omega)^{n+1}}$.

Therefore,

$$\begin{aligned} 10F[te^{-2t}u(t)] &= 10 \left[\frac{1!}{(2 + j\omega)^{1+1}} \right] \\ &= 10 \left[\frac{1}{(2 + j\omega)^2} \right] \\ &= \frac{10}{(2 + j\omega)^2} \end{aligned}$$

Thus, the Fourier transform of the function $f(t)$ is $\boxed{\frac{10}{(2 + j\omega)^2}}$.

Step 4 of 5

(c)

The function is,

$$p(t) = t[u(t) - u(t-2)]$$

The Fourier Transform of the function $p(t)$ is,

$$F[p(t)] = 75\{F[tu(t)] - F[tu(t-2)]\}$$

The expression is written as,

$$\begin{aligned} P(\omega) &= 75\{F[tu(t)] - F[(t-2+2)u(t-2)]\} \\ &= 75\{F[tu(t)] - F[(t-2)u(t-2)] - F[2u(t-2)]\} \end{aligned}$$

By using time shifting and frequency differentiation properties of the Fourier Transform,

$$f(t-a) \xleftrightarrow{F} e^{-j\omega a} F(\omega) \text{ and } t^n f(t) \xleftrightarrow{F} (j)^n \frac{d^n}{d\omega^n} F(\omega) \text{ respectively,}$$

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The Fourier Transform of the function $p(t)$ is,

$$\begin{aligned} P(\omega) &= 75 \left\{ \frac{1}{(j\omega)^2} - \frac{1 \times e^{-j2\omega}}{(j\omega)^2} - 2 \frac{1 \times e^{-j2\omega}}{j\omega} \right\} \\ &= 75 \left\{ \frac{-1}{\omega^2} + \frac{e^{-j2\omega}}{\omega^2} + \frac{2je^{-j2\omega}}{\omega} \right\} \\ &= \frac{75(e^{-j2\omega} - 1)}{\omega^2} + \frac{150j}{\omega} e^{-j2\omega} \end{aligned}$$

Thus, the Fourier Transform of the function $p(t)$ is $\boxed{\frac{75(e^{-j2\omega} - 1)}{\omega^2} + \frac{150j}{\omega} e^{-j2\omega}}$.